

Zhijie Zheng

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Github | Google Scholar

EDUCATION

University of California, Davis

Ph.D. in Computer Science (Advisor: Dr. Jiawei Zhang) | GPA: 3.9/4.0

Davis, CA
Sept 2025 - June 2030 (Expected)

Sun Yat-sen University

M.S. in Electronic Science | GPA: 3.9/4.0

Guangzhou, China
June 2025

Sun Yat-sen University

B.S. in Electronic Science | GPA: 3.9/4.0

Guangzhou, China
June 2022

SELECTED PUBLICATIONS

8 papers, 80+ citations on Google Scholar

1. PRISM: Feed-Forward Single-Image 3D Reconstruction via Geometric Warp-Residual Modeling.

Z. Zheng, X. Xiang, J. Zhang. Under Review at WACV 2027.

Paper link

2. TTSA3R: Training-Free Temporal-Spatial Adaptive Persistent State for Streaming 3D Reconstruction.

Z. Zheng, X. Xiang, J. Zhang. Under Review at WACV 2027.

Paper link Project github

3. UniPT: A Unified Representation Pre-Training for Multi-dataset 3D Object Detection.

Z. Zheng*, C. Wang*, et al. (*equal contribution) PRICAI 2025.

Paper link Project github

INTERNSHIP EXPERIENCE

Beijing Novauto Technology | Research Intern, LiDAR 3D Perception

March 2024 - Oct 2024

- Built UniPT to let one 3D detector train on merged Waymo, nuScenes and KITTI data by learning a shared representation.
- Pre-trained with masked point reconstruction, occupancy and semantic supervision, and cross-dataset feature alignment.
- Outperformed Uni3D (CVPR 2023) by 2.29 mAPH on Waymo, 1.45 BEV mAP on nuScenes and 2.08 3D mAP on KITTI.

RESEARCH EXPERIENCE

RDGSplat: Novel View Synthesis from Frozen 3D Foundation Models | UC Davis

July 2026 - Present

- Diagnosed why 3D foundation models render novel views poorly: their geometry targets metric accuracy, not rendering.
- Engineered render decoders and a pose-conditioned adapter on a frozen 1.4B model, leaving depth and pose intact.
- Raised PSNR by 2.0 to 3.4 dB on four benchmarks with gains on all three backbones tested, training 14.6% of parameters.

PRISM: Feed-Forward Single-Image 3D Reconstruction | UC Davis

Feb 2026 - June 2026

- Identified that warping the input image covers up to 90% of a target view, leaving little for diffusion to generate.
- Replaced diffusion sampling with a lightweight encoder that corrects only the warping residual, trained on synthetic data.
- Cut inference time by 277 $\times$ versus diffusion-based LYRA (2.8 h to 36 s per scene) at comparable quality.

TTSA3R: Streaming 3D Reconstruction for Long Sequences | UC Davis

Oct 2025 - Feb 2026

- Traced pose drift in streaming 3D reconstruction over long videos to uniform overwriting of the recurrent memory state.
- Devised a training-free, plug-and-play state update, gated per token by state evolution and observation quality.
- As sequences grew from 50 to 250 frames, error rose only 1.33 $\times$ versus over 4 $\times$ for CUT3R; cited by 7 works in 8 months.

LiDAR 3D Object Detection for Autonomous Driving | SYSU

Aug 2022 - Dec 2023

- Published 4 journal papers on sparse LiDAR 3D detection for autonomous driving (2 first-author, IEEE TMM and SPL).
- Designed RAFDet, fusing point and range-view features via bidirectional attention, achieving +4.2 mAP on Waymo.
- Proposed DTSSD, a dual-channel PointNet++ and Transformer backbone, achieving +4.18 mAP on KITTI.

TECHNICAL SKILLS

Research Areas: 3D Reconstruction, Novel View Synthesis, 3D Gaussian Splatting, 3D Foundation Models, LiDAR Detection

Programming Languages: Python, C/C++, CUDA, Shell

Deep Learning & CV Frameworks: PyTorch, OpenPCDet, OpenCV, Open3D, Point Cloud Library (PCL)

Tools & Platforms: Linux (Ubuntu), Docker, Git, Robot Operating System (ROS)

Language Skills: Chinese Mandarin (native), English (fluent)

HONORS & AWARDS

National Scholarship for Master's Students in 2023

Dec 2023

Second Prize (South China Division), 2023 China College IC Competition

Aug 2023

Outstanding Undergraduate Thesis Award, Sun Yat-sen University

June 2022